History-Dependent Repair in a Minimal Non-Neural Carrier

A causal test of distributed pattern memory versus path-dependent attractor dynamics

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THE BIOLOGICAL QUESTION

Work on anatomical homeostasis and basal cognition treats morphogenesis as collective problem-solving in morphospace: tissues can recover large-scale organization after perturbation, while physiological networks retain history not reducible to current anatomy alone [1–4]. **Can later repair depend on history retained in the causal organization of the same substrate that conducts repair?**

The contribution is a joint intervention assay. One local-rule carrier must earn closure through body–world contact, retain the resulting relational history after the world is deleted, express it during a later matched injury, and lose the A/B difference when the candidate representation is disrupted. No complete-body target register, reward, or online selector participates in the modeled loop.

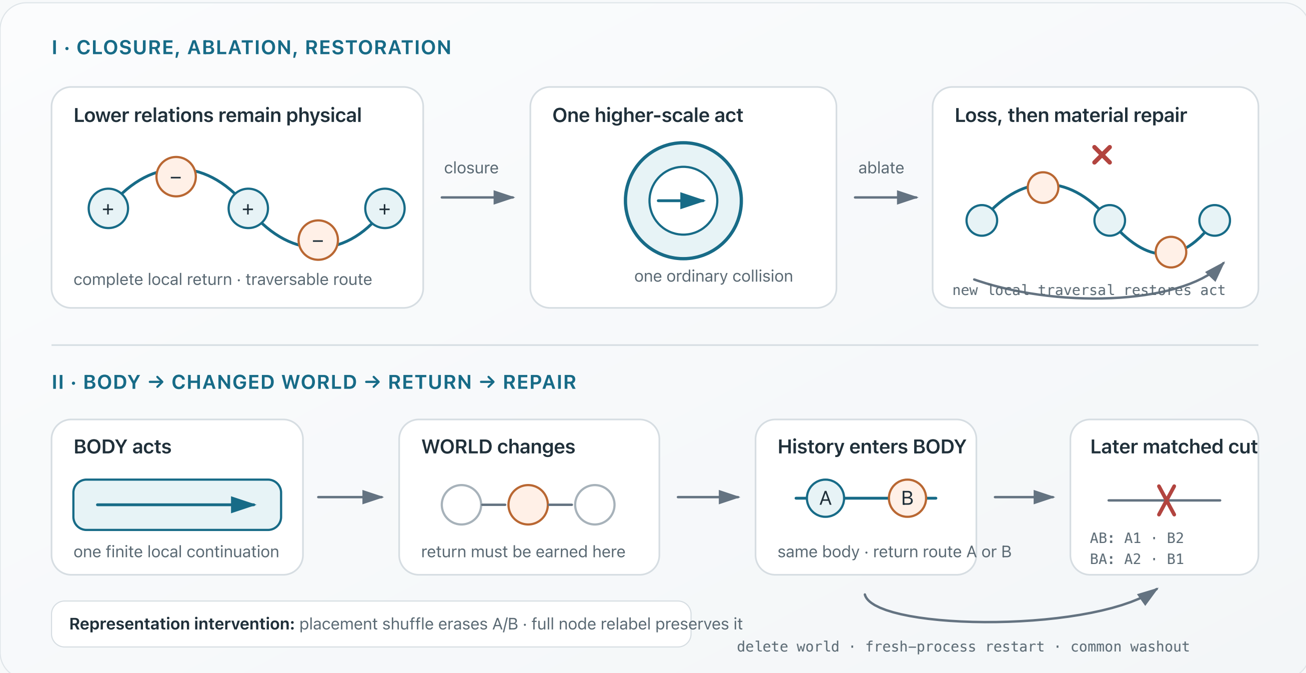

Figure 1. The carrier joins two intervention courts: an upper relation earned by prior closure must disappear under ablation and return through the surviving lower body; an earned body–world history must survive world deletion, fresh-process restart, common washout, and renewed injury.

CAUSAL ASSAY

A history claim must change future conduct under intervention—not merely classify a trace.

The harness fixes the A/B return opportunity before execution; it does not select a relation online.

1. BODY action changes a particular WORLD record.
2. Return can emerge only from that changed record.
3. The original world is deleted; BODY alone restarts.
4. Common washout and the same new injury follow.
5. Relabel must preserve A/B; placement shuffle must erase it.

1 rule

collision kernel SHA-256
`0d32047eecb3…`

THE CARRIER

A deterministic body stores 1,024 writable records with explicit local relation handles. One rule updates one linked encounter at a time. The harness prepares topology, injury, worlds, and readout; A/B placement is fixed before the run. There is no global sweep, coordinate geometry, target register, reward, planner, or score-dependent update. The five-record world uses the same digital physics as BODY—an explicit limitation. Frozen audit: 21 tests · world-lineage source SHA-256: `2ac3304f0a95…`

History-dependent repair

A and B start from the same injured 513-record body. Equal-resource worlds differ only in where an earned final return enters two prepared routes. Both recover 1,024 records and the same current upper conductance. After world deletion, fresh-process restart, identical washout, and the same cut, A restores the upper entry in episode 1 of AB and 2 of BA; B gives the mirror pattern.

Interventions on the representation

Full address relabeling with relations preserved retains body organization and future A/B behavior. Shuffling all 257 candidate return-relation placements erases A/B while preserving their multiset, material extent, current upper conductance, and other relation classes. Separately, ablating the upper relation after closure changes one-collision PASS to HALT; one traversal of the surviving 12-place cycle rebuilds PASS.

A1 · B2
upper-entry birth episode in AB; mirrored in BA

4 / 4
prespecified deterministic initializations; not independent replicates

112
paired G1/G2 world-continuity controls

8 + 8
placement-shuffle + full-relabel interventions

What the result supports

A bounded constructive result: past world contact is retained as changed relational placement in this body, and that placement causally changes later repair after world deletion, restart, common washout, and renewed injury. The interventions identify a distribution-dependent relation class, not a small memory locus.

What remains unresolved

Topology, closure affordance, routes, injury, and readout are engineered. The prepared organization may itself be a rudimentary distributed target or only a path-dependent attractor. This is not a biological model, independent replication, evidence of basal cognition, or evidence that morphogenesis lacks target states.

QUESTION FOR BIOLOGY

Would you regard the topology-encoded closure affordance and history-bearing relational state as a **minimal distributed pattern memory or homeostatic target, only a path-dependent attractor, or as both?** What single perturbation would best distinguish their causal roles?

REFERENCES

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